ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/2
PAPER 2 November 2018
2 hours 30 minutes
PAPER 2 — STRUCTURED QUESTIONS — CALCULATOR ALLOWED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2018 4004/2.

Additional materials: Mathematical tables or a non-programmable calculator may be used.
INSTRUCTIONS: Section A: answer all questions (52 marks). Section B: answer any four of the seven questions (48 marks).

SECTION A (52 marks) — Answer ALL questions

1. (a) Make t the subject of $v = u + at$. [3]<<
>>(b) Simplify $(3x/4) + (x/6)$. [3]<<
>>(c) Solve $2/x = 5/(15)$. [4] **[10]**

2. 29 Form 4 pupils at Fletcher High scored:<<
>>6 (7 pupils), 3 (7 pupils), 2 (5 pupils), 4 (5 pupils), 3 (5 pupils).<<
>>(a) Modal mark. [1]<<
>>(b) Mean mark. [4]<<
>>(c) Range. [2]<<
>>(d) One extra pupil scores 6. Effect on the mean? [2] **[9]**

3. AB ∥ CD with transversal EF in a street plan of Bulawayo.<<
>>(a) Angle APQ = 59°. State the corresponding angle at Q, with a reason. [2]<<
>>(b) Find the co-interior angle to 59°. [2]<<
>>(c) A triangle has angles 59°, 44° and x°. Find x. [2]<<
>>(d) Classify the triangle (acute / right / obtuse). [2] **[8]**

4. (a) Using ruler and compasses only, construct triangle ABC with AB = 8 cm, BC = 7 cm and AC = 9 cm. [4]<<
>>(b) Construct the perpendicular bisector of BC. [2]<<
>>(c) Mark the circumcentre and describe its property. [2] **[8]**

5. A cylindrical tank at Fletcher High has radius 7 cm and height 20 cm. $\pi = 22/7$.<<
>>(a) Volume. [3]<<
>>(b) Curved surface area. [3]<<
>>(c) Total surface area including the base only (open top). [2] **[8]**

6. Line L: $y = 3x + 5$.<<
>>(a) Gradient and y-intercept. [2]<<
>>(b) Coordinates where L meets the x-axis. [2]<<
>>(c) Show A(0, 5) lies on L. [2]<<
>>(d) Equation of the line parallel to L through (0, 4). [2] **[8]**

SECTION B (48 marks) — Answer any FOUR of the seven questions

7. OA = a = (2 ; 1), OB = b = (-1 ; 5). M midpoint of AB.<<
>>(a) Vector AB. [3]<<
>>(b) |AB|. [3]<<
>>(c) Position vector OM. [3]<<
>>(d) Show AM = MB. [3] **[12]**

8. A grain silo is observed from A on level ground in Bulawayo. Angle of elevation of the top is 30°. Distance A to the foot is 24 m.<<
>>(a) Height of the object. Use tan 30° = $1/\sqrt{3}$. [4]<<
>>(b) From B, 24 m due East of A, the bearing of the foot is 300°. Sketch. [3]<<
>>(c) A bird sits halfway up. Angle of elevation from A. [5] **[12]**

9. (a) Solve $x^2 - 9x + 18 = 0$ by factorisation. [4]<<
>>(b) A rectangular plot in Bulawayo is 6 m longer than it is wide. If the area is 27 m^2, form and solve an equation for the width w. [5]<<
>>(c) Sketch $y = (x - 3)(x - 6)$, showing intercepts. [3] **[12]**

10. O is the centre. A, B, C on the circumference. AB is a diameter. Angle BAC = 36°.<<
>>(a) Angle ACB. Reason. [3]<<
>>(b) Angle BOC (centre, same arc BC). [3]<<
>>(c) D on the remaining circumference. Angle BDC. Reason. [3]<<
>>(d) State the angle-in-a-semicircle theorem. [3] **[12]**

11. (a) y varies directly as x. When x = 4, y = 48. Find y when x = 10. [4]<<
>>(b) z varies inversely as x. When x = 3, z = 8. Find z when x = 6. [4]<<
>>(c) A farmer near Bulawayo has at most 40 hours. Maize takes 3 h/ha, vegetables 1 h/ha. Inequality for hectares m and v. [4] **[12]**

12. Triangle P(1,2), Q(4,2), R(1,4).<<
>>(a) Reflect in the y-axis. Coordinates of P'. [4]<<
>>(b) Translate PQR by (3 ; 1). Give Q''. [3]<<
>>(c) Matrix (0 −1 ; 1 0) maps (x;y). Describe fully and find the image of Q. [5] **[12]**

13. 80 pupils at Fletcher High sat a test (0–50). An ogive is drawn.<<
>>(a) How to read the median from the ogive. [3]<<
>>(b) Median estimated as 22. Meaning? [2]<<
>>(c) How to find the IQR from the graph. [4]<<
>>(d) Why IQR is better than the range here. [3] **[12]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).